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Review

Pharmacological activities and therapeutic potential of Hyperoside in the treatment of Alzheimer's and Parkinson's diseases: A systemic review

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ABSTRACT

Alzheimer's disease (AD) and Parkinson's disease (PD) are neurodegenerative disorders that significantly impact well-being. Hyperoside (HYP), a flavonoid found in various plant species, particularly within the genus Hypericin, exhibits diverse pharmacological properties. However, the precise mechanisms underlying the anti-AD and anti-PD effects of HYP remain unclear. This systematic review consolidated existing preclinical research on HYP by conducting a comprehensive literature survey and analysis. The objective was to corroborate the therapeutic efficacy of HYP in AD and PD models and to synthesize its potential therapeutic mechanisms. Searches were conducted in the PubMed, CNKI, and Web of Science databases. Reliability assessment of the 17 included studies confirmed the credibility of the mechanisms of action of HYP against AD and PD. We systematically assessed the neuroprotective potential of HYP in in vivo and in vitro models of AD and PD. Our findings indicated that HYP can mitigate, intervene in, and treat AD and PD animal models and associated cells through various mechanisms, including anti-oxidative, anti-inflammatory, anti-apoptotic, anti-Aβ aggregation, and cholinesterase inhibitory activities. Therefore, HYP potentially exerts anti-AD and anti-PD effects through diverse mechanisms, making it a promising candidate for therapeutic intervention in both AD and PD.

Introduction

Alzheimer's disease (AD), the most prevalent neurological degenerative disease globally, stands as the principal contributor to dementia (Mazon et al., 2017). With the continuous increase in the percentage of the elderly population worldwide, the incidence of AD has consistently risen, evolving into a significant global public health concern (Takizawa et al., 2015). Broadly characterized as a progressive deterioration of cognitive and behavioral functions within the central nervous system, AD manifests as a degenerative affliction (Gao et al., 1998). Notably, the hallmark pathological manifestations of AD encompass the anomalous accumulation of amyloid β (Aβ) plaques within neuronal structures alongside the formation of neurofibrillary tangles attributed to hyperphosphorylated tau protein, thereby precipitating neuronal malfunction (Porsteinsson et al., 2021). During the early phases of AD, astrocytes engulf a considerable volume of Aβ42 fibrils, wherein the incomplete degradation of these fibrils culminates in the release of neurotoxic agents by astrocytes, thereby inciting neuronal detriment (Davidson, 2009). Tau, a pivotal microtubule-associated protein, plays a crucial role in stabilizing neuronal microtubules and exhibits high expression levels within the central nervous system (Muralidar et al., 2020). The stability of Tau protein is intricately regulated by a variety of kinases, and any aberrant phosphorylation of Tau can lead to disrupted microtubule architecture, thereby initiating neuronal functional impairment (Wegmann et al., 2021). Currently, two classes of pharmacotherapeutic agents have received approval for the treatment of AD. The first class comprises acetylcholinesterase inhibitors, such as donepezil and galantamine (Pellegrini et al., 2021), which elevate cerebral acetylcholine levels by inhibiting cholinesterase activity (Sharma, 2019). These agents are typically prescribed for individuals with mild to moderate AD. The second class includes N-methyl-D-aspartate (NMDA) antagonists, exemplified by memantine, which are indicated for individuals with moderate to severe AD (Anand et al., 2017). Additionally, recent advancements have introduced two anti-Aβ monoclonal antibodies, aducanumab and lecanemab, as potential contributors to AD treatment. However, these medications primarily ameliorate AD symptoms without halting disease progression. Consequently, further research and development efforts aimed at identifying novel therapeutic agents for AD are

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imperative.

Parkinson's disease (PD) ranks as the second most prevalent neurological degenerative disorder globally, imposing significant physical and economic burdens on aging populations. Its incidence has witnessed a marked increase over the past three decades and is projected to persist over the next three decades (Chen et al., 2020). The primary clinical presentations of PD include static tremors, bradykinesia, myotonia, and postural gait disturbances. Additionally, non-motor symptoms such as rapid eye movement, sleep disorders, depression, and anosmia progressively emerge with disease progression, typically manifesting in advanced stages (Jankovic and Tan, 2020). As PD advances, both motor and non-motor symptoms exacerbate, significantly impairing the quality of life of the patients and imposing substantial burdens on their families and society at large. The predominant etiology of PD is currently attributed to the depletion of a significant population of dopamine neurons within the substantia nigra pars compacta (Kalia and Lang, 2015). Despite extensive research, the precise etiology and pathogenesis of PD remain elusive. Various factors including genetic predisposition, aging, pesticide exposure, and environmental toxins have been implicated in PD pathogenesis (Elsworth, 2020). Moreover, emerging evidence suggests that mitochondrial dysfunction, oxidative stress, and inflammatory processes contribute to the pathophysiology of PD (Chatterjee et al., 2020). Current therapeutic strategies for PD primarily aim at replenishing dopamine levels or utilizing dopamine receptor agonists such as metoba and pramexol. However, these approaches primarily address symptom alleviation and lack interventions to halt disease progression. Thus, there exists an urgent need for the continual development of innovative therapeutics for PD.

Hyperoside (HYP), also known as hyperin or quercetin-3-O-galactoside, is a flavonoid glycoside abundantly present in various plant species, including members of the *Hypericaceae*, *Rosaceae*, and *Platycodon* families (Wang et al., 2023) (Fig. 1). Its pharmacological profile encompasses anti-inflammatory, antioxidant, antidepressant, and vasculoprotective properties (Wang et al., 2022), rendering it applicable across diverse pathological conditions such as cancer, diabetes, depression, and cardiovascular diseases (Zhang et al., 2020; Hu et al., 2020; Song et al., 2022; Yang et al., 2021). HYP exhibits the capability to attenuate reactive oxygen species (ROS) overproduction and preserve islet β cell function, thereby contributing to the prevention and management of type 2 diabetes mellitus (Chen et al., 2022). Furthermore, consumption of flavonoid-rich foods has been associated with a reduced risk of specific vascular diseases and cancers (Ivey et al., 2017). Proposed mechanisms include modulation of the oxLDL-LOX-1-ERK pathway, suppression of VCAM-1 upregulation in TNF-α-mediated vascular smooth muscle cells, inhibition of the NFκB signaling cascade, and regulation of intracellular calcium signaling (Zhang et al., 2017; Jang et al., 2018; Qiu et al., 2019; Jia et al., 2020). The potential role of HYP in neurological disorders has garnered attention, particularly in traditional Chinese medicine where it has demonstrated efficacy in alleviating depression and cerebrovascular ailments (Nguyen et al., 2024). Consequently, researchers have been motivated to explore its prospective therapeutic application in neurodegenerative conditions due to its multifaceted effects. However, the comprehensive protective mechanisms of HYP against neurological disorders remain incompletely understood. This study provides a review of recent experimental investigations involving HYP in *in vitro* and *in vivo* models, offering detailed insights into its mechanisms of action within the nervous system.

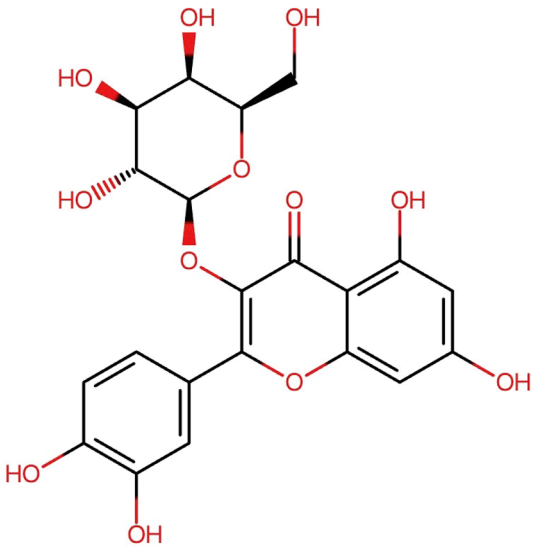

Fig. 1. The structure of Hyperoside.

Methods

The systematic review was designed and executed in adherence to the guidelines provided by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement. However, PROSPERO is still under registration.

Search strategy

A systematic exploration was conducted across PubMed, CNKI, and Web of Science databases, covering the literature from inception to January 2024, encompassing both Chinese and English publications, to identify studies regarding Hyperoside (HYP) and its interactions with the nervous system. The search strategy involved combining HYP with specific keywords such as "nervous system," "neurodegenerative diseases," and "biomarkers," along with particular neurodegenerative conditions like "Alzheimer's disease" and "Parkinson's disease". Titles and abstracts underwent screening to identify pertinent articles, followed by a comprehensive review of the selected studies.

Inclusion and exclusion criteria

The following inclusion criteria, based on the PICO principle, were used to identify appropriate studies: (1) participants: AD and PD model animals and cells; (2) intervention: Hyperoside; (3) comparison: the control group comprised untreated control, vehicle control and positive drug control; (4) outcomes: improve cognitive dysfunction in AD models; alleviate the loss of dopamine neurons in PD models and neuroprotection both in AD and PD models.

The exclusion criteria for the studies in this systematic review were as follows: (1) literature reviews, (2) investigations involving the concurrent administration of HYP with other therapeutic agents, and (3) studies lacking a control group.

Data Extraction and treatment evaluation

The retrieved literature was imported into EndNote, where duplicate documents were systematically identified and eliminated. Two researchers (Y.J.Y and D.X.Y) independently conducted literature searches based on inclusion and exclusion criteria. The initial phase of the review involved a meticulous screening of titles and abstracts to filter out studies that were not pertinent to the research question at hand. Following this preliminary screen, a more in-depth review was conducted by thoroughly examining the full text of the remaining articles. The following data were extracted from each study: (1) species utilized for in vivo investigations and cell types employed for in vitro analyses, (2) pharmaceutical agents and dosages administered in in vitro and in vivo experiments, (3) dosage regimen and duration of HYP treatment, (4) publication year and primary authorship of the study, and (5) principal findings and underlying mechanisms elucidated in the

investigation.

Quality assessment

The risk of bias was evaluated using the SYRCLE (Systematic Review Centre for Laboratory Animal Experimentation) tool for animal studies. This comprehensive evaluation encompassed ten distinct criteria: sequence generation, baseline characteristics, allocation concealment, random housing, blinding of experimentalists, random outcome assessment, blinding of outcome assessors, incomplete outcome data, selective outcome reporting and other sources of bias. Each study was assigned a maximum possible score of 10 points. Two independent assessors, Y.J.Y and Z.S.Y, conducted a rigorous and autonomous examination of the risk of bias within the studies that were included in the review. Any discrepancies arising from the quality assessments were ultimately reconciled through a collaborative consultation process involving the corresponding author. In total, SYRCLE tool was used to conduct bias risk assessment for 8 literatures (Table 1), and the scores of most of the articles were 5 or 6, indicating that the quality of the included literatures was at a medium or high level. Tabe;The methodological qualities of the 8 studies are delineated in this review (Fig. 2).

Results

Study selection

Upon initial screening, 87 studies were identified, comprising 45 from PubMed, 30 from CNKI, and 12 from the Web of Science. After resolving database duplications and applying exclusion and inclusion criteria, 22 articles were deemed eligible. Subsequently, the abstracts of these 22 articles were examined, resulting in the inclusion of 17 articles for this review (Fig. 3).

Quality assessment

A comprehensive evaluation of methodological quality was conducted on the selected 8 articles using the CAMARADES tool (Table 2). The assessment scores based on the CAMARADES checklist are provided in Table 2. Most studies achieved a score of five to seven, indicating moderate-to-high quality.

Table 1
Methodological quality of included studies based on SYRCLE.

Study	A	B	C	D	E	F	G	H	I	J	Total
Yi et al., 2022	+	+	?	+	?	+	?	?	+	+	6
Chen et al., 2021	+	+	?	+	?	+	?	?	+	+	6
Nam et al., 2014	+	+	?	−	?	+	?	?	+	+	5
Wang et al., 2023	−	+	?	+	?	+	?	+	+	+	6
Song et al., 2023	+	+	?	+	?	+	?	?	+	+	6
Wang et al., 2022	?	+	?	+	?	+	?	?	+	+	5
Xu et al., 2023	+	+	?	+	?	+	?	?	+	+	6
Fan et al., 2021	+	+	?	−	?	+	?	?	+	+	5

A: Sequence generation; B: Baseline characteristics; C: Allocation concealment; D: Random housing; E: Blinding of experimentalists; F: Random outcome assessment; G: Blinding of outcome assessors; H: Incomplete outcome data; I: Selective outcome reporting; J: Other sources of bias. "+" indicates low risk, "-" indicates high risk, "?" reflects "unclear".

Study characteristics

This review encompasses 17 studies focusing on AD (n = 12) and PD (n = 5).

AD

Three *in vivo* studies (Yi et al., 2022; Chen et al., 2021; Nam and Lee, 2014) were incorporated into the compilation and organization of AD data utilizing species such as APPswe, PSEN1dE9 (APP/PS1) mice (n = 1), imprinting control region mice (n = 1), and CD-1 mice (n = 1). Among these studies, one employed Aβ1-42-induced AD models, one utilized APP/PS1 mice models, and another study employed scopolamine 1-induced AD models. Across the three *in vivo* investigations, administered doses of HYP ranged from 10–250 mg, and the duration of HYP intervention ranged from 3–9 months. Behavioral assessments, including the Morris water maze, passive avoidance, and object recognition tests, were employed to evaluate therapeutic efficacy. Further details of these studies are provided in Table 3.

Moreover, this review encompasses seven *in vitro* studies (Huang et al., 2021; Xin et al., 2019; Zeng et al., 2011; Liu et al., 2018; Kraus et al., 2007; Liu et al., 2005), utilizing various cell types including PC12 cells (n = 2), HT22 cells (n = 1), primary astrocytes (n = 1), primary cultured rat neurons (n = 1), bEnd.3 cells (n = 1), and BV2 and N11 cells (n = 1). These investigations induced AD utilizing Aβ, hydrogen peroxide (H_2O_2), lipopolysaccharide (LPS), glutamate, and sodium azide. Administered doses of HYP across the seven *in vitro* studies ranged from 2.5–500 μM, 10–180 μg, or 0.01–1 μmol, with treatment durations ranging from 8 to 24 h.

Furthermore, this review includes two studies encompassing both *in vivo* and *in vitro* models (Wang et al., 2023; Song et al., 2023) employing *Caenorhabditis elegans*, APP/PS1 mice, PC12 cells, and HT22 cells. These investigations utilized the CL4176 strain and Aβ to induce AD, with treatment durations spanning from 24 h to 8 weeks. Additional details are provided in Tables 3–6.

PD

In the context of PD, this review incorporated two *in vitro* studies (Fan et al., 2017; Kwon et al., 2019), employing BV2 microglial cells and SH-SY5Y cells (n = 1), as well as SH-SY5Y cells (n = 1). These studies used LPS and 6-hydroxydopamine to induce PD. Administered doses of HYP ranged from 0.1 to 20 μM, with treatment durations varying from 24 to 36 h.

Additionally, three studies involving both *in vivo* and *in vitro* models (Wang et al., 2022; Xu et al., 2023; Fan et al., 2021) were included in this review. C57BL/6J (n = 1), C57BL/6 (n = 1), and SD (n = 1) mice were used as experimental animals. SH-SY5Y (n = 2), PC12, and BV-2 cells (n = 1 each) were used for these experiments. *In vivo* PD induction was achieved using 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine and rotenone, whereas *in vitro* PD induction was achieved using LPS, 1-methyl-4-phenylpyridinium ion, and rotenone. Administered doses of HYP ranged from 10–200 μg, 25–150 μg, and 0.25–2 μM across these *in vivo* and *in vitro* studies, with treatment durations ranging from 3 h to 28 days. Further details are provided in Tables 7-10.

Summary of the studies

AD

In vivo studies

Yi et al. (Yi et al., 2022) employed CD-1 mice to induce nervous system damage via Aβ1-42 administration. Aβ1-42 induces mitochondrial dysfunction, leading to cognitive impairment in murine models (Wang et al., 2014). In the Aβ model, Aβ1-42 was infused into the lateral ventricle, followed by treatment with HYP (10/30/100 mg/kg/d) once

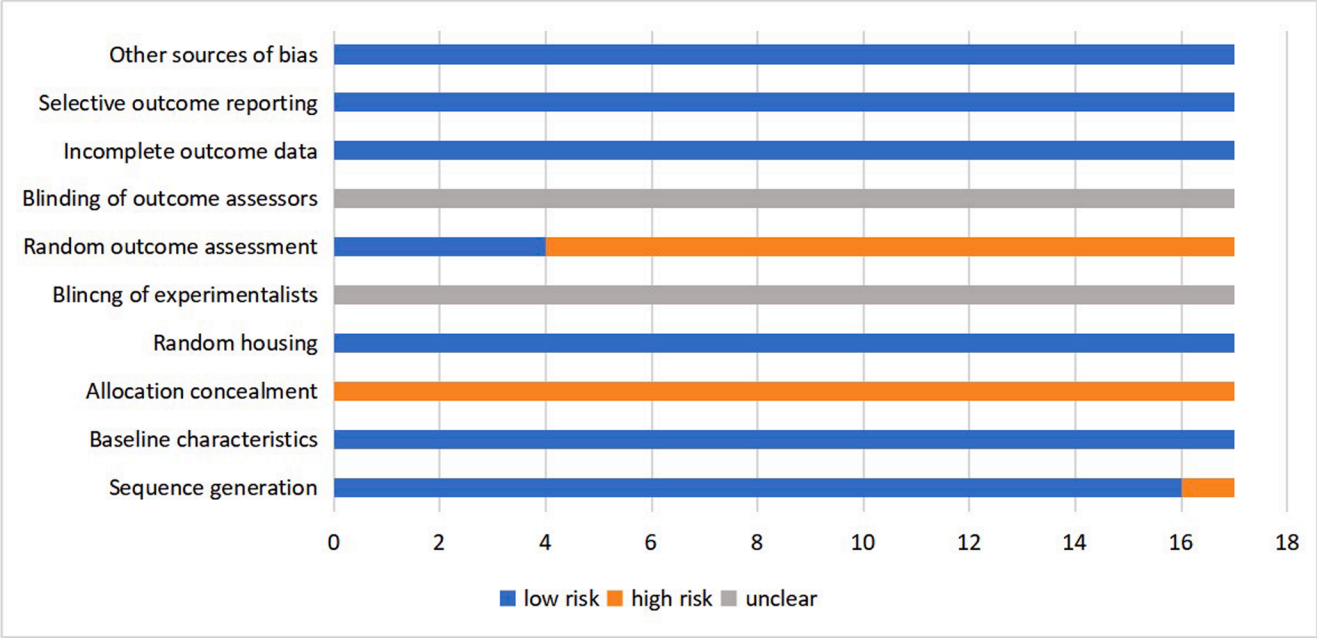

Fig. 2. Risk of bias graph.

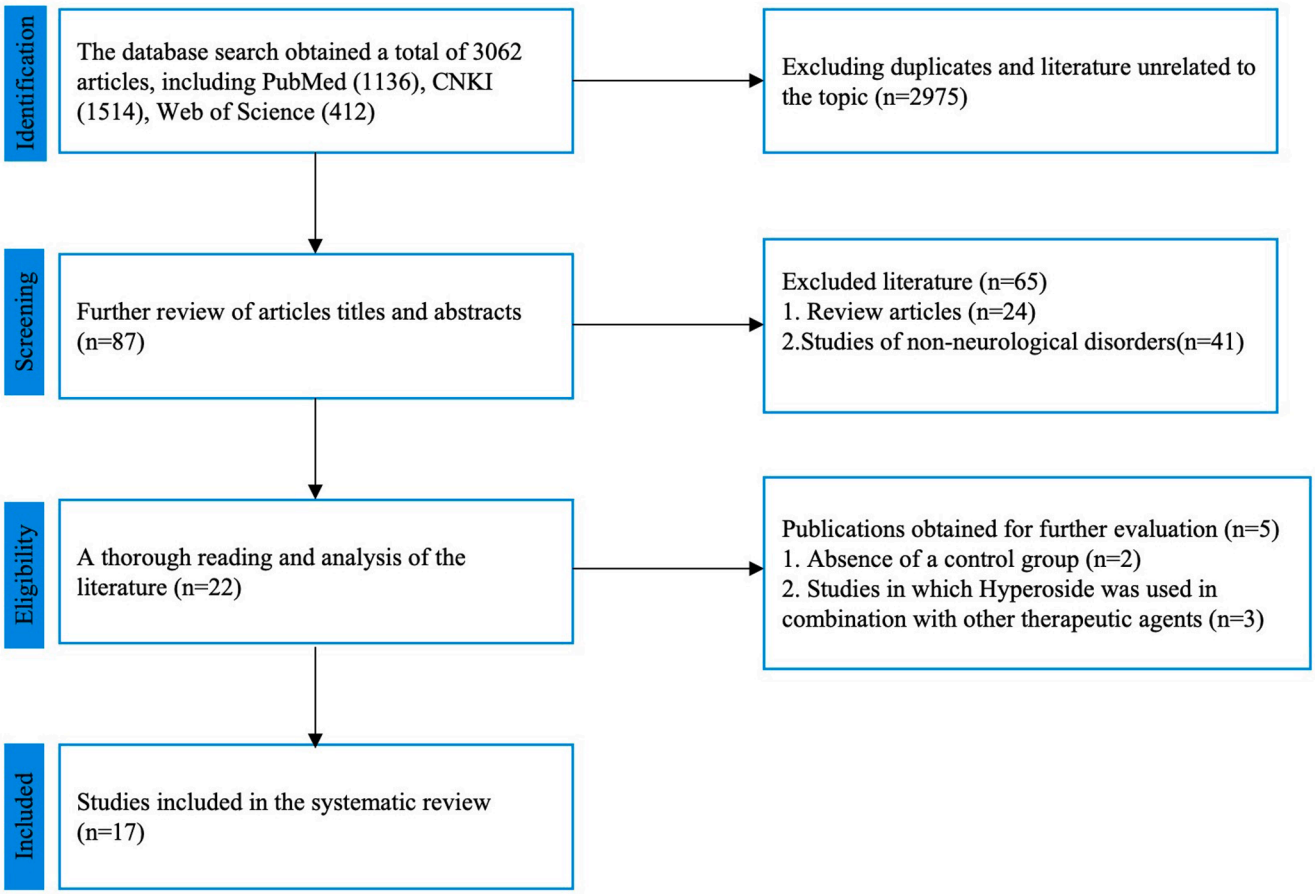

Fig. 3. PRISMA flow chart of search strategy and study selection process.

daily for 3 days. Behavioral assays were performed to assess the therapeutic effects of HYP. Notably, in the passive avoidance test, HYP significantly prolonged the dark avoidance latency in a dose-dependent manner, suggesting enhanced learning and memory. Furthermore, in the Aβ-induced cognitive decline model, HYP ameliorated Aβ-induced memory deficits by upregulating hippocampal long-term potentiation

Table 2
Quality Assessment of Included Studies based on CAMARADES checklists.

Study	A	B	C	D	E	F	G	H	I	J	Total
Yi et al., 2022	√	√	–	–	√	√	√	√	–	–	6
Chen et al., 2021	√	√	–	–	√	√	√	√	√	–	7
Nam et al., 2014	√	–	–	–	√	√	√	√	–	–	5
Wang et al., 2023	√	√	–	–	√	√	√	√	–	–	6
Song et al., 2023	√	√	–	–	√	√	√	√	–	–	6
Wang et al., 2022	√	√	–	–	√	√	√	√	–	–	6
Xu et al., 2023	√	–	–	–	√	√	√	√	–	–	5
Fan et al., 2021	√	√	–	–	√	√	√	√	–	–	6

A: peer-reviewed journal; B: animals were randomly allocated; C: blinded outcome assessment; D: calculation of sample size; E: statement of compliance with animal welfare regulations; F: possible conflicts of interest; G: appropriate animal model; H: temperature control; I: blind established model; J: reporting of animals excluded from analysis.

Table 3
Studies that investigated the use of Hyperoside in vivo models for Alzheimer's disease.

Study (years)	Species (n, Sex)	Model	Interventions	Research methods
Yi et al., 2022	CD-1 mice (n = 50, Male)	Aβ1-42 (10 μM, 3 μl)	HYP (10/30/100 mg/kg/d) for 3 days	Passive avoidance test; Object recognition test
Chen et al., 2021	APP/PS1 double transgenic mice (n = 36, Half male and half female)	APP/PS1 mice	HYP (50 mg/kg/d) for 9 months	MWM
Nam et al., 2014	ICR mice (n = 21, Male)	SCOP 1 mg/kg	HYP (250 mg/kg/d) for 7 days	Passive avoidance test; MWM
Wang et al., 2023	Caenorhabditis elegans	CL4176 strain and CL2355 strain	HYP (40/80/120 μM) for 36 h	Learning ability assay
Song et al., 2023	APP/PS1 mice (n = 43, Male)	APP/PS1 mice	HYP (20/40/80 mg/kg/3days) for 8 weeks	Y-maze tests; Fear-conditioning tests; Rotarod test

Abbreviation: Aβ, β-amyloid; APP/PS1, APPswe, PSEN1dE9; CL, Caenorhabditis elegans; HYP, Hyperoside; MWM, Morris Water Maze; SCOP1, scopolamine 1.

(LTP) via the activation of calcium-permeable amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid receptors (CP-AMPARs). HYP can reduce the fibrous Aβ plaques and positive diffuse deposition in APP/PS1 mice models. Chen et al. (Chen et al., 2021) administered APP/PS1 mice with HYP (50 mg/kg/d) for 9 months to determine its effects on improving cognition. They examined the learning ability of mice with AD, mainly spatial learning and memory, using the Morris Water Maze (MWM) method (Bromley-Brits et al., 2011), and suggested that HYP can significantly improve cognitive impairment. In addition, HYP reduced the number of activated microglia and astrocytes, thereby exerting anti-inflammatory and antioxidant effects by alleviating β-secretase (BACE1) and glycogen synthase kinase-3β levels. To explore the relationship between HYP and AD, Nam et al. used scopolamine to establish a mouse model of memory impairment in vivo (Nam and Lee, 2014). Scopolamine is a non-selective competitive muscarinic receptor inhibitor (Anacker, 2018) that induces cognitive impairment in animal models (Xie et al., 2023). It affects neurotransmission in the central nervous system and significantly reduces reactivity and neurotransmission in the hippocampal cholinergic system, thereby causing learning and memory impairment (Choi et al., 2021). HYP (250 mg/kg/day) was administered to each mouse for 7 days. Behavioral tests included passive avoidance and MWM task tests. They revealed that HYP can ameliorate memory impairment at a dose of 250 mg/kg due to its anti-acetylcholinesterase and anti-oxidant activities.

Table 4
Main outcome and mechanism of Hyperoside in vivo models for Alzheimer's disease.

Study (years)	Results	Mechanism
Yi et al., 2022	Improved learning ability and memory impairment	Increase hippocampal LTP through CP-AMPAR activation
Chen et al., 2021	Improved spatial learning and memory; Reduced amyloid plaque deposition and tau phosphorylation, decreased the number of activated microglia and astrocytes, and attenuated neuroinflammation and oxidative stress	Antioxidant activity; Reduce BACE1 and GSK3 β levels
Nam et al., 2014	Ameliorated memory impairment	Anti-acetylcholinesterase; Antioxidant activities
Wang et al., 2023	Reduced ROS level and Aβ42-induced neurotoxicity in C. elegans worms	Antioxidant ability
Song et al., 2023	Improved motor deficit, spatial memory and learning ability; Reduced Aβ plaques; Attenuated apoptosis in the cortex and hippocampus of mice	Anti-Aβ aggregation; Attenuate apoptosis

Abbreviation: Aβ, β-amyloid; BACE1, β-secretase; CP-AMPAR, calcium-permeable amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid receptor; GSK-3β, glycogen synthase kinase-3β; LTP, long-term potentiation; ROS, reactive oxygen species.

In summary, HYP exerts a protective effect on the nervous system in vivo. HYP can improve cognitive impairment in AD animals by increasing the levels of acetylcholine and activating CP-AMPARs, which play antioxidant and anti-inflammatory roles. Therefore, HYP has great potential as a monomeric drug for treating nervous system diseases.

In vitro studies

Seven in vitro studies scrutinized the mechanistic underpinnings of HYP action in AD. One study examined four standard nerve cell types in antioxidant experiments conducted in vitro, including SH-SY5Y, HT22, PC22, and primary astrocytes. Xin et al. (Xin et al., 2019) observed that HYP mitigated the glutamate-induced ROS production, implying robust antioxidant efficacy and neuroprotective benefits in these cellular models. Two in vitro studies used PC12 cell models to determine the pharmacological effects of HYP on neurodegenerative diseases. Li et al. and Liu et al. (Liu et al., 2005) used sodium azide, H_2O_2, and *tert*-butyl hydroperoxide to induce cell apoptosis. Li et al. reported that HYP attenuated sodium azide-induced apoptosis in PC12 cells. Moreover, Liu et al. observed that HYP prevented the PC12 cells from shrinking and undergoing apoptosis induced by H_2O_2 and *tert*-butyl hydroperoxide. HYP significantly promoted the growth of PC12 cells in a dose-dependent manner. In summary, HYP exhibited antioxidant and anti-apoptosis activities. Huang et al. (Huang et al., 2021) observed that HYP suppresses apoptosis, inflammation, and oxidative stress elicited by LPS in HT22 cells by upregulating silent mating type information regulation 2 homolog-1, thereby triggering the activation of the Wnt/β-catenin and sonic hedgehog pathways. Three in vitro studies used Aβ to build cellular models of cognitive impairment. Zeng et al. (Zeng et al., 2011) reported that HYP inhibited the apoptotic cascade of caspase cells and maintained the mitochondrial integrity, thereby inhibiting Aβ25-35-induced cytotoxicity and apoptosis. In bEnd.3 cells, Liu et al. (Liu

Table 5
Studies that investigated the use of Hyperoside in vitro models for Alzheimer's disease.

Study (years)	Species (n, Sex)	Model	Interventions	Research methods
Huang et al., 2021	HT22 cells	LPS (1 μg/mL)	HYP (2.5/5/10/20 μM) for 24 h	MTT, WB, qRT-PCR
Xin et al., 2019	PC22, SH-SY5Y, HT22 and primary astrocytes	Glutamate (8.8 mM, 67 mM, 6.9 mM and 5.1 mM)	HYP (64 μg/ml) for 24 h	ABTS assay; DPPH assay
Zeng et al., 2011	Primary cultured rat neurons	Aβ25–35 (20 μM)	HYP (2.5/5/10/20 μM) for 24 h	MMP; WB; Co-immunoprecipitation assay
Liu et al., 2018	bEnd.3 cells	Aβ1-42 (20 μM)	HYP (50/200/500 μM) for 24 h	MTT; WB
Kraus et al., 2007	BV2 cells and N11 cells	Aβ fragments (25–35), (35–25) and (1–40)	HYP (75 μg/ml) for 24 h	MTT assay; LDH-release assay
Li et al., 2011	PC12 cells	Sodium azide (20 mmol/L)	HYP (0.01/0.1/1 μmol/L) for 24 h	MTT; WB; CM-H2DCFDA assay
Liu et al., 2005	PC12 cells	H_2O_2 (150 μM), t-BuOOH (300 μM)	HYP (10/20/40/60/80/100/120/140/160/180 μg/ml) for 8 h	HPLC assay; Flow cytometric detection; LDH activity assays
Wang et al., 2023	PC12 cells	Aβ42 (2.5 μM)	HYP (10/20/30 μM) for 24 h	WB
Song et al., 2023	HT22 cells	Aβ (30 μM)	HYP (20–80 μM) for 24 h	WB

Abbreviation: Aβ, β-amyloid; ABTS, 2,2′-azino-bis-(3-ethylbenzothiazoline6-sulphonic acid); CM-H2DCFDA, 5-(and-6)-chloromethyl-2′, 7′dichlorodihydrofluorescein diacetate; DPPH, 2,2-diphenyl-1-picrylhydrazyl; H_2O_2, hydrogen peroxide; HYP, Hyperoside; LDH, lactate dehydrogenase; LPS, Lipopolysaccharide; MMP, mitochondrial membrane potential; MTT, 3-(4,5-dimethylthiazol-2-yl)-2,5diphenyltetrazolium bromide; t-BuOOH, *tert*-butyl hydroperoxide; WB, Western blotting.

et al., 2018) revealed that HYP could inhibit the hyperpermeability of the blood–brain barrier induced by cellulose Aβ1-42, thus playing an anti-apoptotic role in neuronal cells. Finally, Kraus et al. (Kraus et al., 2007) reported that HYP reduced the microglial cell viability by altering the cell membrane fluidity.

In conclusion, the reported therapeutic effect of HYP on the nervous system provides further possibilities for its clinical use for the treatment of nervous system diseases.

In vivo and in vitro studies

Two studies examined the effects of HYP on AD in vivo and in vitro. ROS is an important index for measuring oxidative stress in vivo. Aβ induces oxidative stress in Caenorhabditis elegans, resulting in the development of AD. Wang et al. (Wang et al., 2023) reported that HYP reduced the ROS levels in vivo and improved the learning ability of Caenorhabditis elegans models. In vitro studies confirmed that HYP increased the Aβ42-induced mitochondrial membrane potential of PC12 cells and decreased the ROS levels through exerting its antioxidant effects. Another in vitro and in vivo study explored the mechanism by which HYP ameliorates neuronal death. Song et al. reported that HYP can improve the learning memory and motor disorders in an in vivo model of APP/PS1 mice. In addition, HYP reduced the aggregation of Aβ fibers and apoptosis of the cortex and hippocampus in mice, thereby exerting antioxidant and anti-apoptotic effects. Similarly, in HT22 cells,

Table 6
Main outcome and mechanism of Hyperoside in vitro models for Alzheimer's disease.

Study (years)	Results	Mechanism
Huang et al., 2021	Inhibited the LPS-induced apoptosis, inflammation and oxidative stress production	Increase STRI to activate the Wnt/β-catenin and sonic hedgehog pathways
Xin et al., 2019	Attenuated glutamate induced ROS production	Antioxidant ability
Zeng et al., 2011	Inhibited Aβ25–35-induced cytotoxicity and apoptosis	Maintain mitochondrial integrity; Inhibit caspase apoptotic cascade
Liu et al., 2018	Inhibited blood–brain barrier hyperpermeability induced by fibrillar Aβ1-42	Anti-apoptosis
Kraus et al., 2007	Reduced viability of microglial cells	Alter cellular membrane fluidity
Li et al., 2011	Attenuated sodium azide-induced apoptosis in PC12 cells	Anti-apoptosis activity
Liu et al., 2005	Prevented the PC12 cells from shrinking and turning against apoptosis induced by hydrogen peroxide and *tert*-butyl hydroperoxide	Antioxidant activity; Anti-apoptosis activity
Wang et al., 2023	Increased mitochondrial membrane potentialin Aβ42-induced PC12 cells	Antioxidant ability
Song et al., 2023	Attenuated Aβ −induced apoptosis	Anti-Aβ aggregation Attenuate apoptosis

Abbreviation: Aβ, β-amyloid; LPS, Lipopolysaccharide; ROS, reactive oxygen species; STRI, silent mating type information regulation 2 homolog-1.

Table 7
Studies that investigated the use of Hyperoside in vitro models for Parkinson's disease.

Study (years)	Species (n, Sex)	Model	Interventions	Research methods
Fan et al., 2017	The BV2 microglial cells and SH-SY5Y cells	LPS (100 ng/mL)	HYP (2.5, 5, 10 and 20 μM) for 24 h	MTT; WB
Kwon et al., 2019	SH-SY5Y Cells	6-OHDA (200 μM)	HYP (0.1, 0.25, 0.5, 1 and 2 μM) for 24 h	MTT; WB; TUNEL Assay

Abbreviation: 6-OHDA, 6-hydroxydopamine; Aβ, β-amyloid; LPS, Lipopolysaccharide; HYP, Hyperoside; MTT, 3-(4,5-dimethylthiazol-2-yl)-2,5diphenyltetrazolium bromide; WB, Western blotting.

Table 8
Main outcome and mechanism of Hyperoside in vitro models for Parkinson's disease.

Study (years)	Results	Mechanism
Fan et al., 2017	Inhibited LPS-induced microglia activation through p38 and NF-κB pathways	Anti-inflammatory activity
Kwon et al., 2019	Exhibited significant neuroprotection against 6-OHDA-induced oxidative stress and neurotoxicity through activating the Nrf2-mediated HO-1 transcriptional signaling pathway	Ani-oxidative stress; Neuroprotective effects

Abbreviation: 6-OHDA, 6-hydroxydopamine; HO-1, heme oxygenase-1; HYP, Hyperoside; LPS, Lipopolysaccharide; Nrf2, the nuclear erythroid 2-related factor 2.

Table 9
Studies that investigated the use of Hyperoside *in both* in vivo *and* in vitro models for Parkinson's disease.

Study (years)	Species (n, Sex)	Model	Interventions	Research methods
Wang et al., 2022	C57BL/6J mice (n = 60, Male); PC12 cells and BV-2 cells	In vivo: MPTP (30 mg/kg) In vitro: LPS (1 mg/mL)	In vivo: HYP (10, 20, 40 mg/kg/d) for 3 days In vitro: HYP (10, 20, 40 mg/kg/d) for 3 h	Rotarod test; Pole test; Open field test and swimming test
Xu et al., 2023	C57BL/6 mice (n = 40, Male); SH-SY5Y cells	In vivo: MPTP In vitro: MPP + (2 mM)	In vivo: HYP (25 mg/kg) for 7/14 days In vitro: HYP (25, 50, 75, 100, 125 and 150 μg/mL) for 24 h	Gait test
Fan et al., 2021	SD mice (n = 40, Male); SH-SY5Y cells	In vivo: Rotenone (2.5 mg/kg) In vitro: Rotenone (1 μM)	In vivo: HYP (100, 200 mg/kg/d) for 28 days In vitro: HYP (0.25, 0.5, 1 and 2 μM) for 24 h	Rotarod test

Abbreviation: HYP, Hyperoside; LPS, Lipopolysaccharide; MPP+, metabolized to 1-methyl-4-phenylpyridinium ion; MPTP, N-methyl-4-phenyl-1,2,3,6-tetrahydropyridine; SD, Sprague–Dawley.

Table 10
Main outcome and mechanism of Hyperoside *in both* in vivo *and* in vitro models for Parkinson's disease.

Study (years)	Results	Mechanism
Wang et al., 2022	Alleviated motor dysfunction; attenuated degeneration of dopaminergic neurons and inhibits activation of SNpc neuroglial cells in mice with Parkinson's disease model; Inhibited activation of NLRP3 inflammatory vesicles via PACAP-CREB pathway	Anti-inflammation
Xu et al., 2023	Attenuated behavioral deficits and dopaminergic neuron loss in mice with Parkinson's disease; Reduced oxidative stress and mitochondrial damage and induced secretion of neurotrophic factors	Antioxidant activity; Activate Akt pathway
Fan et al., 2021	Ameliorated the behavioral deficits in rats; Overturned the increase in cell apoptosis and the loss of cell mitochondrial membrane potential induced by rotenone	Inhibit cell apoptosis

Abbreviation: LPS, Lipopolysaccharide; NLRP3, NLR Family Pyrin Domain Containing 3; PACAP, pituitary adenylate cyclase-activated peptide; SNpc, substantia nigra compacta.

HYP blocked the destruction of the Aβ-induced mitochondrial membrane potential (MMP) in HT22 cells and exerted anti-apoptotic and brain protective effects (Song et al., 2023).

In conclusion, both in vivo models and in vitro investigations present compelling medical evidence for the pharmacological efficacy of HYP, indicating its potential as a therapeutic agent for AD. HYP exhibits diverse biological actions including inhibition of Aβ aggregation, anti-oxidative properties, and anti-apoptotic effects, all of which are conducive to the advancement of AD therapeutics.

PD

In vitro studies

Two in vitro studies investigated the pharmacological effects of HYP on PD, where SH-SY5Y cells were used. Fan et al. (Fan et al., 2017) used LPS to induce the activation of BV2 microglial cells and SH-SY5Y cells, and they constructed a PD model based on neuroinflammation. HYP exerted anti-inflammatory effects by inhibiting LPS-induced microglial activation through the p38 and NFκB pathways. Moreover, Kwon et al. (Kwon et al., 2019) reported that HYP has a significant neuroprotective effect against 6-hydroxydopamine-induced oxidative stress and neurotoxicity by activating the Nrf2-mediated HO-1 (heme oxygenase-1) transcriptional signaling pathway.

These findings suggest that HYP plays a potential role in the prevention and treatment of PD, and further in-depth studies are needed. In addition, this in vitro study provided a scientific basis for the development of HYP as a potential therapeutic agent for PD.

In vivo and in vitro studies

Three in vivo and in vitro studies were conducted on HYP in PD, and these studies reviewed the underlying mechanisms of HYP in PD. One study explored the mechanism by which HYP protects dopaminergic (DA) neurons. Wang et al. (Wang et al., 2022) constructed an in vivo C57BL/6J mouse model. Behavioral experiments, including the rotarod, pole, open field, and swimming tests, were used to verify the active effects of the drug. HYP alleviated the motor dysfunction in the mice models. In addition, HYP reduced the degeneration of DA neurons and inhibited the activation of substantia nigra pars compacta glial cells. PC12 and BV-2 cells were used for in vitro experiments, where HYP exerted anti-inflammatory effects by inhibiting the activation of the NLR family pyrin domain-containing 3 (NLRP3) inflammasome through the pituitary adenylate cyclase-activated peptide (PACAP)-CREB pathway. These studies suggest that HYP has considerable benefits for the treatment of dyskinesia and inflammation caused by PD. Xu et al. (Xu et al., 2023), who established in vivo models using C57BL/6 mice and in vitro models using SH-SY5Y cells, revealed that HYP alleviated the behavioral disorders and the loss of DA neurons in mice with PD. In addition, HYP reduced the oxidative stress and mitochondrial damage and induced the secretion of neurotrophic factors, thereby protecting against the loss of DA neurons. In summary, HYP activates the Akt pathway. Fan et al. (Fan et al., 2021) used rotenone to establish in vivo and in vitro PD models. Rotenone is a toxin that inhibits the mitochondrial electron transport chain complex I (Innos and Hickey, 2021). This can cause the asymptotic loss of DA neurons to construct a Rotenone-induced PD model (Fan et al., 2021). Fan et al. reported that HYP can improve the behavioral deficits in rats and inhibit the rotenone-induced potentiation of apoptosis.

In conclusion, HYP can reduce neuroinflammation and damage caused by PD through its anti-apoptosis, antioxidant, and anti-inflammatory effects. Although many studies have been conducted, a deeper discussion is still needed to fully evaluate the efficacy of HYP and expand its clinical application as an effective drug for the treatment of PD.

Discussion

HYP, a flavonoid glycoside derived from the *Hypericum* genus, predominantly sourced from the fruits or herbs of *Rhododendroceae* and *Hypericaceae* (Chunzhi et al., 2016), exhibits a spectrum of pharmacological effects, including anti-inflammatory, antioxidant, anti-cancer, and antidepressant properties (Qiu et al., 2019). Nevertheless, the precise mechanisms underlying the therapeutic potential of HYP in neurodegenerative disorders, such as AD and PD, remain partially understood. Upon the analysis of our findings, we discerned that HYP possesses the capacity to forestall, intervene, and treat AD and PD in animal models and relevant cellular systems through multifaceted mechanisms, including anti-oxidative, anti-inflammatory, anti-apoptotic, anti-Aβ aggregation, and cholinesterase inhibitory activities (Figs. 4 and 5).

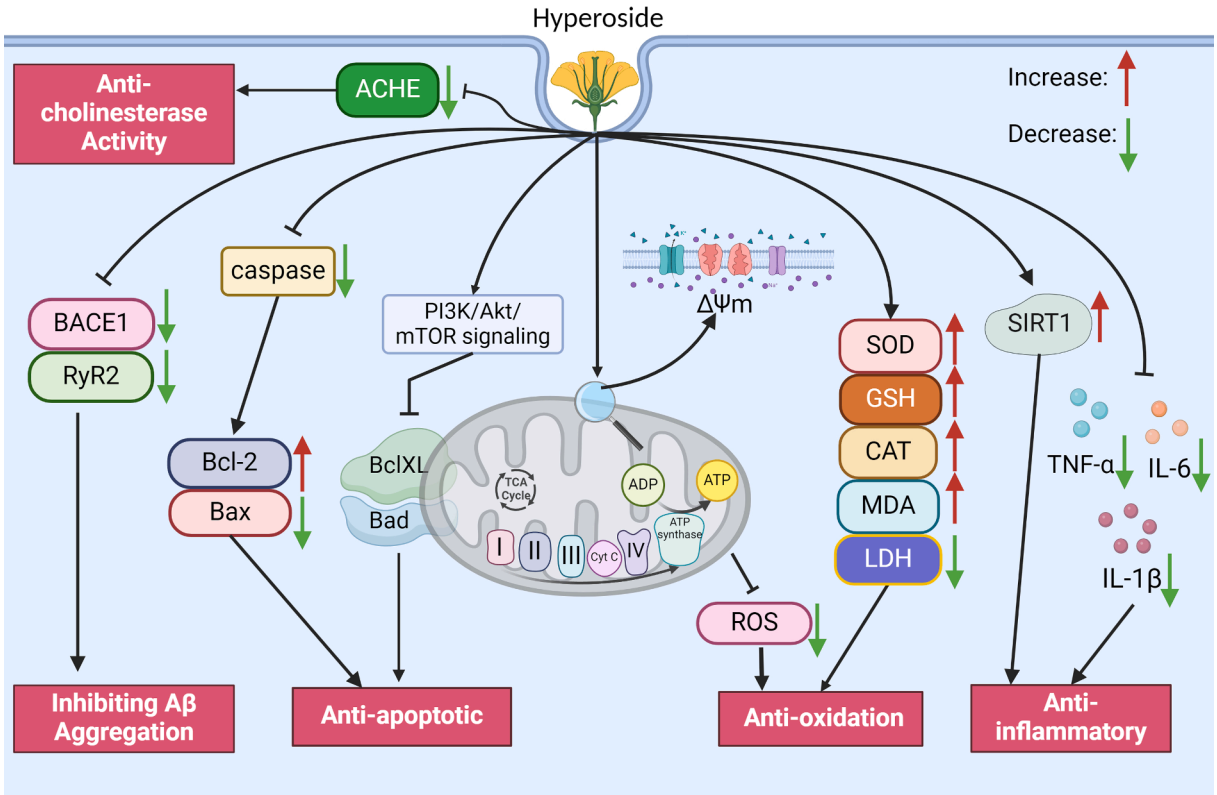

Fig. 4. The neuroprotective mechanism of Hyperoside (HYP) in Alzheimer's disease. HYP inhibits ROS production by increasing MMP levels and enhances anti-oxidant activity through the upregulation of SOD, GSH, CAT, and MDA. HYP treatment elevates SIRT1 levels and reduces pro-inflammatory factors such as IL-1β, IL-6, and TNF-α, mediating its anti-inflammatory effects. HYP prevents Bad from binding to BclXL by activating the PI3K/Akt/mTOR signaling pathway, increases Bcl-2 levels, and decreases Bax levels, contributing to its anti-apoptotic effects. HYP inhibits BACE1 and RyR2 levels to reduce Aβ aggregation and lowers acetylcholinesterase levels, thereby exerting its anti-cholinesterase activity. Abbreviations: Aβ, β-amyloid; ACHE, acetylcholinesterase; AKT, protein kinase B; BACE1, β-secretase; CAT, catalase; Cyt c, cytochrome *c*; GSH, glutathione; HYP, Hyperoside; IL, interleukin; LDH, lactate dehydrogenase; MDA, malondialdehyde; MMP, mitochondrial membrane potential; ROS, reactive oxygen species; RyR, ryanodine receptor; SIRT1, Sirtuin 1; SOD, superoxide dismutase; TNF-α, tumor necrosis factor-α.

Anti-oxidation

Oxidative stress (OS) refers to oxidative damage caused by an oxidative imbalance in the body or cells. It is involved in the pathogenesis of AD and is closely associated with ROS production (Tang et al., 2024). An imbalance in OS can produce excessive ROS, which exceed the antioxidant capacity of the body and cause various neurological diseases (Orfali et al., 2024). OS is characterized by ROS overproduction, damage to the mitochondrial respiratory chain, and calcium imbalance (Park et al., 2021). Impaired mitochondrial metabolism is a major cause of neurological degeneration (Golpich et al., 2017). Mitochondrial membrane potential (ΔΨm) is an important reference index to evaluate the mitochondrial function status (Samanta et al., 2024). HYP significantly increased the levels of MMP (ΔΨm), decreased the levels of ROS, and played a role in OS. Moreover, HYP demonstrated the capacity to significantly enhance the activities of key antioxidant enzymes within the cellular milieu, including superoxide dismutase (SOD), glutathione peroxidase (GSH-Px), and catalase (CAT). Malondialdehyde (MDA) is a commonly used marker for lipid peroxidation and is pivotal in assessing oxidative stress levels. HYP exerts protective effects by elevating the MDA levels (Wang et al., 2023). Related studies have shown that HYP can reduce the ROS levels through the cell membrane by chelating excess metal ions in the cell, and plays an anti-oxidative role. Extracellular lactate dehydrogenase levels can be used to evaluate the antioxidant properties of the therapeutic drugs. *Liu et al.* noted that HYP reduced the extracellular lactate dehydrogenase levels, providing another crucial evidence for its antioxidant activity (Liu et al., 2005). In addition, HYP scavenges free radicals, and its antioxidant capacity is derived from the presence of partial polyphenol residues in its molecular structure and the reduction of oxidative stress kinase levels (Xin et al., 2019). Chen *et al.* reported that HYP plays an anti-oxidative role by increasing the levels of SOD and decreasing the expression of MDA, the end product of lipid peroxidation, to improve learning and memory and prevent AD (Chen et al., 2021).

The mechanism of OS in DA neurons plays an important role in the pathogenesis of PD. OS can cause degeneration and loss of function of DA neurons, thus causing the occurrence of PD (Mohan et al., 2024). Under physiological conditions, cellular oxidation and antioxidant reactions exist in a dynamic balance (Tan and Suda, 2018). When the body is exposed to external stimuli, homeostasis in the cell is destroyed and excessive accumulation of ROS results in the inactivation of functional proteins and production of lipid peroxidation products, ultimately leading to neurodegenerative changes (Newsholme et al., 2016). Nuclear erythroid 2-related factor 2 (Nrf2) is a pivotal transcription factor involved in anti-oxidative stress mechanisms. Heme oxygenase-1 (HO-1) acts as a downstream mediator of Nrf2 and operates concurrently with Nrf2 to modulate cellular responses to oxidative stress (Dai et al., 2022). HYP significantly activates the antioxidant factor Nrf2 and induces the overexpression of HO-1, which plays an antioxidant role (Kwon et al., 2019). Mitochondria is the main site of aerobic respiration and provides energy for the basic cellular activities. The oxidative respiratory chain in the inner mitochondrial membrane is involved in important physiological functions including ROS production, calcium homeostasis, and energy supply (Harvey, 2019). Mitochondrial dysfunction can cause excessive ROS production and oxidative stress, resulting in neuronal dysfunction. Xu *et al.* reported that HYP can increase the MMP, and thus play a protective role in mitochondrial dysfunction. Brain-derived neurotrophic factor, a glial-derived neurotrophic factor, reverses

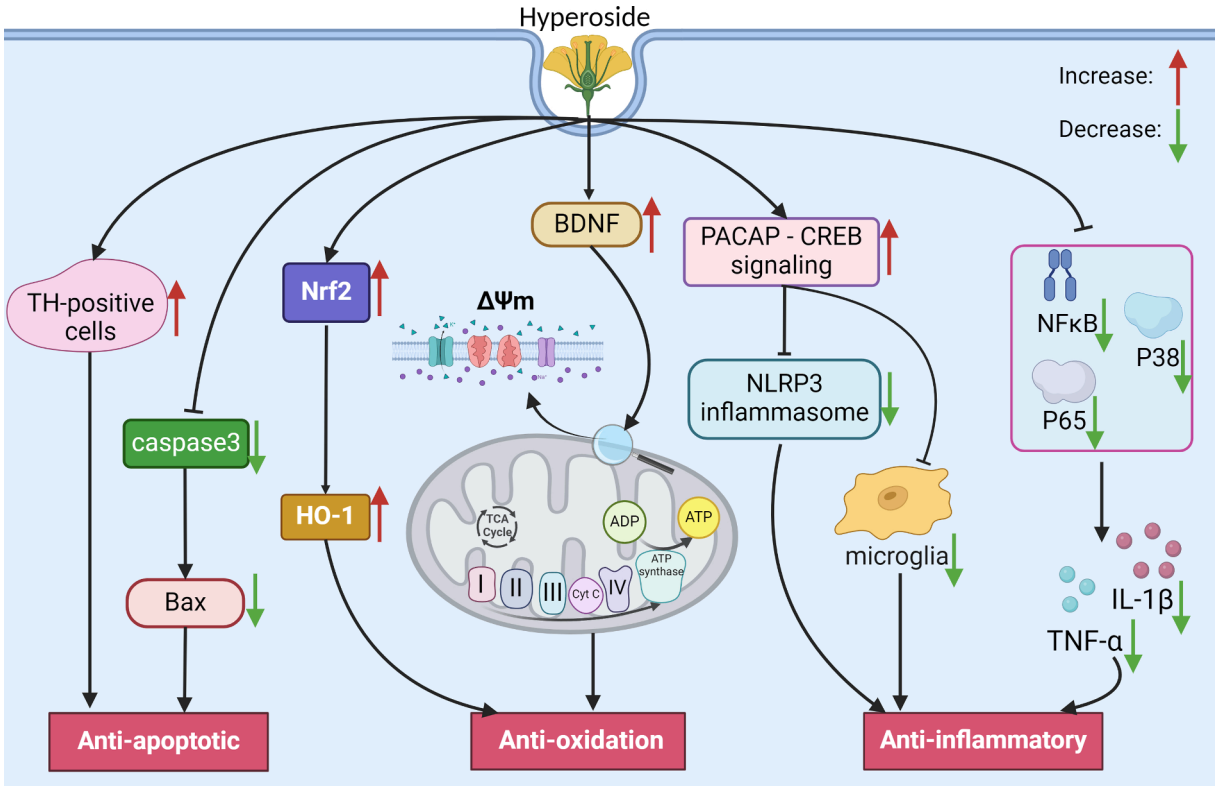

Fig. 5. The neuroprotective mechanism of Hyperoside (HYP) in Parkinson's disease. HYP increases the number of TH-positive cells and reduces Bax levels through caspase, exerting its anti-apoptotic effects. HYP enhances antioxidant activity by increasing Nrf2, HO-1, and BDNF levels. It inhibits the activation of the NLRP3 inflammasome and microglia by activating the endogenous PACAP-CREB pathway, thereby exerting anti-inflammatory and neuroprotective effects. HYP reduces the production of IL-1β and TNF-α by inhibiting the activation of p38 and p65/NFκB. Abbreviations: BDNF, brain-derived neurotrophic factor; HO-1, heme oxygenase-1; HYP, Hyperoside; IL, interleukin; NLRP3, NLR Family Pyrin Domain Containing 3; Nrf2, nuclear factor erythroid 2-related factor 2; PACAP, pituitary adenylate cyclase-activated peptide; TH, tyrosine hydroxylase; TNF-α, tumor necrosis factor-α; ΔΨm, mitochondrial membrane potential.

mitochondrial dysfunction and exerts antioxidant effects (Xu et al., 2023).

Anti-inflammatory

Recent advancements in our understanding of neurodegenerative disorders have underscored the pivotal role of neuroinflammation in the etiology and progression of neurological conditions, such as AD. While the inflammatory response serves as a protective mechanism against external stimuli, its excessive or prolonged activation can transition into chronic inflammation (Leuti et al., 2020; Morales and FARíAS G, MACCIONI R B. , 2010). Chronic inflammation within the central nervous system induces damage to both the tissue and blood–brain barrier, exacerbating the trajectory of AD (Islam et al., 2022). This cascade primarily involves the activation of hyperactive glial cells in the brain, namely microglia and astrocytes, which generate and release various cytokines to orchestrate the inflammatory cascade (Li et al., 2020). Consequently, mitigating the glial cell population in the brain has emerged as an effective strategy for AD treatment (Chen et al., 2021). Notably, during the inflammatory response, diverse pro-inflammatory mediators are mobilized, including interleukin-1β (IL-1β), interleukin-6 (IL-6), interleukin-8 (IL-8), tumor necrosis factor-α (TNF-α), malondialdehyde (MDA), and nitric oxide (NO). HYP exerts an anti-inflammatory effect by reducing the expression levels of pertinent pro-inflammatory mediators including tumor necrosis factor-alpha (TNF-α), interleukin-1β (IL-1β), interleukin-6 (IL-6), and interleukin-8 (IL-8). Sirtuin 1 (SIRT1), a member of the mammalian sirtuin family, is known for its anti-aging and neuroprotective effects (Yan et al., 2019). Moreover, the involvement of the SIRT1 pathway in synaptic plasticity and its neuroprotective role against Aβ aggregation have also garnered attention (Porquet and GRIñáN-FERRé C, FERRER I, , 2014). Huang *et al.* revealed that HYP can enhance the SIRT1 expression levels, thereby activating the Wnt/β-catenin and Sonic hedgehog pathways, thereby exerting an anti-inflammatory effect (Huang et al., 2021).

Inflammation leads to the dysfunction of DA neurons in the substantia nigra. A substantial body of evidence suggests a robust correlation between persistently heightened inflammatory responses and the development of PD. Various proinflammatory cytokines produced during the inflammatory response have been confirmed to play an important role in the development of PD. In addition, NLRP3 inflammasome mediates inflammatory responses. HYP inhibits the activation of the NLRP3 inflammasome and microglia by activating the endogenous PACAP-CREB pathway, which has anti-inflammatory and neuroprotective roles (Wang et al., 2022). In addition, Fan *et al.* reported that HYP can attenuate the activation of microglia and the production of various pro-inflammatory mediators by inhibiting the activation of p38 and p65/NFκB. Moreover, the excessive accumulation of NO has toxic effects on neurons. Finally, HYP inhibits the expression of inducible nitric oxide synthase to reduce the NO release, thereby playing an anti-inflammatory role (Fan et al., 2017).

Anti-apoptotic

Apoptosis, the process of programmed cell death, is pivotal in maintaining physiological homeostasis. The Bcl-2 protein family, known for its anti-apoptotic role, mitigates cell demise (Maiuri et al., 2007). Conversely, Bax, a proapoptotic protein, counteracts this effect. Investigations demonstrate that HYP augments the expression of the anti-apoptotic Bcl-2 protein, while diminishing the Bcl-2/Bax ratio through caspase-3 inhibition. This modulation attenuates the apoptotic cascade,

exerting a notable anti-apoptotic effect. Moreover, mitochondrial dysfunction can induce apoptosis, which is perpetuated by the mitochondria-mediated caspase cascade. Within this framework, Bax facilitates the release of cytochrome *c* (Cyt c) from the mitochondria by enhancing the membrane permeability, resulting in subsequent apoptotic sequelae (Wang et al., 2019). Emerging evidence suggests that HYP can effectively inhibit the apoptotic pathway in mitochondria, thereby protecting the integrity of the blood–brain barrier and reducing the occurrence of AD (Liu et al., 2018). A previous study (Zeng et al., 2011) suggested that HYP exerts anti-apoptotic effects by reversing the mitochondrial dysfunction. Specifically, HYP activates the PI3K/Akt/mTOR signaling pathway, which inhibits the binding of Bad to BclXL, thereby reversing Aβ-induced mitochondrial dysfunction and maintaining the mitochondrial stability.

Although the pathogenesis of PD has not been fully defined, evidence suggests that apoptosis may be an important factor in the loss of DA neurons (Meng et al., 2013). Caspase activation is a key step in apoptosis (Rihova et al., 2024); therefore, the inhibition of caspase activation can exert an anti-apoptotic effect and reverse the neuronal loss. HYP possesses the capacity to reverse the expression levels of caspase 3, Bax, and Bcl-2 in *in vitro* and *in vivo* experimental setup, thereby mitigating the decline in the mitochondrial membrane potential and exerting anti-apoptotic effects. Tyrosine hydroxylase is a rate-limiting enzyme in DA synthesis that increases the number of DA neurons. Fan *et al.* reported that HYP augmented the population of tyrosine hydroxylase-positive cells, ameliorated behavioral abnormalities in rat models, and exhibited an anti-apoptotic function (Fan et al., 2021).

Inhibiting Aβ aggregation

The prevailing "Aβ theory" persists despite the intricate pathogenesis of AD. Dysregulation of the generation and elimination of Aβ peptides within the cerebral milieu has been posited as an inaugural event in AD progression. Aβ, derived from the proteolytic cleavage of β amyloid precursor protein (APP) by β and γ secretases, comprises 39–42 amino acid residues (Dobson, 2017). The accumulation of Aβ within the brain milieu is associated with synaptic and neuronal damage, thereby precipitating AD onset (Huang et al., 2015). Presently, heightened Aβ deposition is implicated in exerting neurotoxic effects on neuronal cells. Consequently, mitigating the cerebral Aβ levels holds promise for the management of AD dementia. The foundational underpinning of learning and memory is synaptic plasticity, which is characterized by the adaptability of synaptic structure and function. The facilitation and sustenance of long-term synaptic plasticity holds promise for enhancing cognitive processes associated with learning and memory (Bazzari and Parri, 2019). HYP has been documented to facilitate acute hippocampal slice synaptic LTP, thereby mitigating Aβ deposition. Moreover, HYP demonstrates the capacity to augments hippocampal LTP through the activation of CP-AMPARs, which is potentially mediated by the activation of adenylyl cyclase and protein kinase A (Yi et al., 2022). Aβ, originating from the stepwise cleavage of APP by BACE1 and γ-secretase, underscores BACE1 as a promising therapeutic target (Tan et al., 2024). BACE1 inhibitors are potential disease-modifying agents used for the treatment of AD (Coimbra et al., 2018). Previous studies have indicated that prolonged HYP therapy substantially reduced the cerebral BACE1 levels and attenuated its enzymatic activity. This attenuation contributes to the preservation of APP integrity, consequently aiding in the reduction of Aβ levels in the organism (Chen et al., 2021).

Calcium ions act as messengers that regulate synaptic function and maintain mitochondrial membrane stability (Endo, 2009). The accumulation of Aβ in the body causes calcium overload, which in turn leads to a reduction in MMP that causes mitochondrial dysfunction. Therefore, the regulation of calcium homeostasis in the body is crucial. The alanine receptor 2 channel is involved in a variety of cellular functions, mediates calcium ion release, and leads to Aβ accumulation. Studies have reported that HYP can reduce alanine receptor 2 and thus Aβ (Song et al., 2023).

The permeability of the cell membrane is essential for maintaining internal and external ion balance. Aβ can form unique channels in the plasma membrane of cells and disrupt the balance of ions and other compounds inside and outside the membrane, thus disrupting the fluidity of the membrane and causing neuronal damage (Bouron, 2024). HYP affects the fluidity of neuronal cell membranes and protects the neuronal cell integrity, thereby reducing the risk of AD (Kraus et al., 2007). In fact, this HYP-mediated effect is closely related to its prominent components (+)-catechin and (−)-epicatechin, which are believed to reduce the toxic effects of amyloid and exert protective effects on the neurons.

Anti-cholinesterase activity

The number and function of cholinergic neurons in the brains of patients with AD are severely reduced, which leads to a series of cognitive dysfunction-related manifestations. Therefore, supplementing acetylcholine or reducing its hydrolysis is a major strategy for treating AD (Mezeiova et al., 2024). Acetylcholinesterase (AChE) and butyrylcholinesterase (BChE) are key enzymes responsible for hydrolyzing the neurotransmitter acetylcholine. Therefore, AChE and BChE inhibitors are key to ameliorating the reduction in the neurotransmitter levels and cognitive function (Kassab et al., 2024). HYP has a significant anti-cholinesterase activity and plays a neuroprotective role (Nam and Lee, 2014).

Many *in vivo* and *in vitro* studies confirmed the potential mechanism of action of HYP against AD and PD, suggesting that HYP may be a reliable drug for the treatment of AD and PD. However, HYP has some limitations in its clinical applications. First, degenerative diseases of the nervous system are very complex and their pathogenesis remains unclear. Therefore, it is difficult to identify drugs with specific mechanisms of action. Second, the current application of HYP in AD and PD mainly focuses on animal and cell experiments, and the quantity and quality of clinical studies are limited. Thus, further in-depth studies are required to verify the efficacy and safety of this treatment strategy. Third, owing to individual differences, specific diseases, drug compatibility, and other related issues, HYP dosage forms and routes of administration need to be further studied to better meet the clinical needs.

Conclusion

HYP, a component derived from traditional Chinese medicine, has demonstrated efficacy in treating various diseases through multiple mechanisms, including reducing oxidative stress, attenuating apoptosis, suppressing inflammatory responses, inhibiting Aβ aggregation, and decreasing acetylcholinesterase activity. Extensive research utilizing various animal and cell models has supported the potential use of HYP in treating AD and PD, positioning it as a candidate for novel therapeutic development in these areas. However, critical aspects such as optimal dosage, administration routes, pharmacokinetics, and comprehensive clinical evaluations remain underexplored. Therefore, further research is imperative to establish the safety and effectiveness of HYP. In conclusion, HYP exhibits potential as a multi-mechanistic anti-AD and anti-PD agent, with promising wide-ranging therapeutic applications in the management of these neurodegenerative disorders.

CRediT authorship contribution statement

Jiayu Yuan: Writing – original draft. **Xiaoyu Dong:** Data curation. **Siyu Zhou:** Formal analysis. **Jianfei Nao:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

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